Inorganic Cume – Inorganic Energy Diagrams

January 17, 1998

Demonstrate your understanding of the attached energy level diagrams by describing the features of the diagrams as much as you can in the time allotted. The following questions should be addressed; other observations are also welcome.

1. Diagram A is related to diagram B. Explain the relationship.

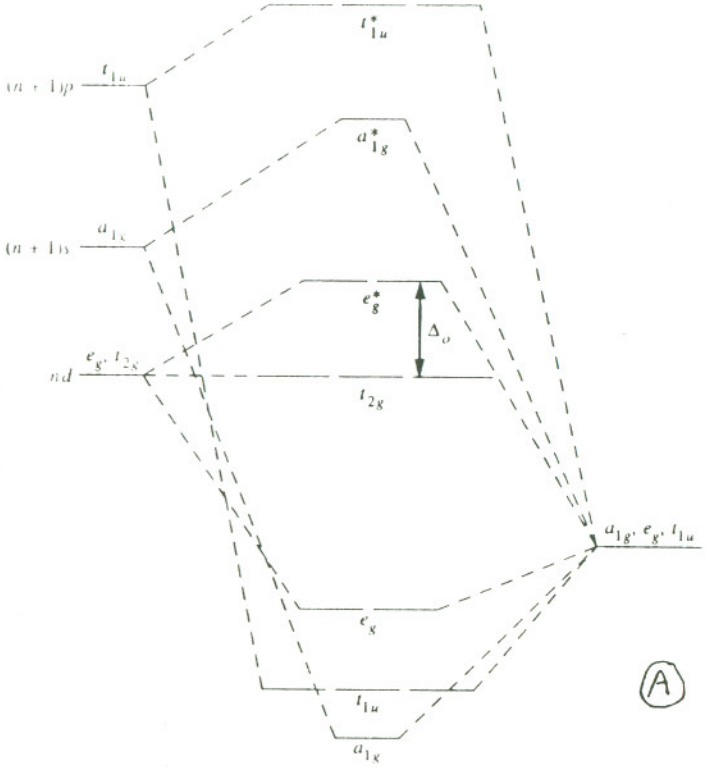
2. Explain (a) what general kind of complex shows the energy splitting pattern B, (b) why the energy splitting occurs, and (c) why the splitting pattern is as shown (why are two up and three down? why are the high levels raised more than the low levels are lowered?).

3. Explain (a) what general kind of complex shows the energy splitting pattern C, (b) why the energy splitting occurs, and (c) why the splitting pattern is as shown (why are three up and two down? why are the high levels raised less than the low levels are lowered?).

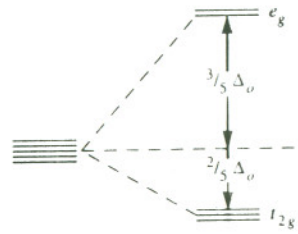
4. Give the energy splitting pattern analogous to B and C for a square-planar complex. Add d orbital labels (d_{xy} etc) to the diagram.

5. The two energy patterns at the left and right sides of diagram D are used to explain complexes of Cu(II). Explain the patterns and tell why they are important for Cu(II).

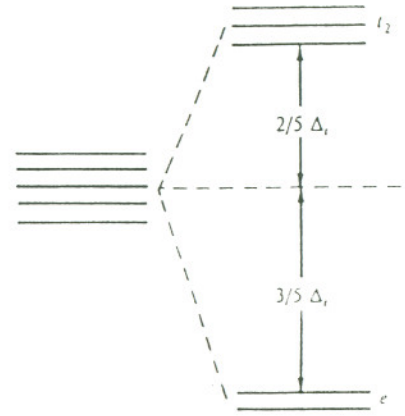
6. The orbital interaction diagrams E and F are related to pi bonding.
   a) Explain why the geometry of this interaction (the orbitals involved) results in an involvement of the t_{2g} levels (and not the e_g levels) of the metal complex.
   b) For diagram E, would the t_{2g} orbitals be filled or empty? (Note the Δ label of the figure) Is this diagram more likely to be used for CO or chloride ion ligands?
   c) For diagram F, would the t_{2g} orbitals be filled or empty? (Note the Δ label of the figure) Is this diagram more likely to be used for CO or chloride ion ligands?
   d) Explain your reasonings for your b and c answers.


t_{1u}^*
$(n+1)p$
t_{1u}
a_{1g}^*
$(n+1)s$
a_{1g}
e_g^*
Δ_o
nd
e_g, t_{2g}
t_{2g}
a_{1g}, e_g, t_{1u}
e_g
t_{1u}
a_{1g}

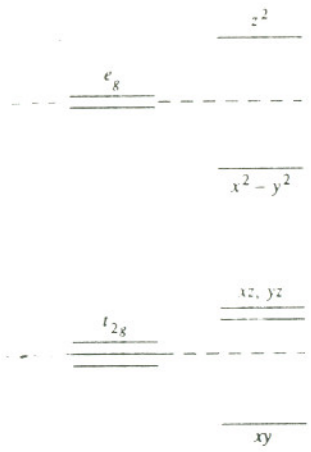
A


e_g
$3/5\ \Delta_o$
$2/5\ \Delta_o$
t_{2g}

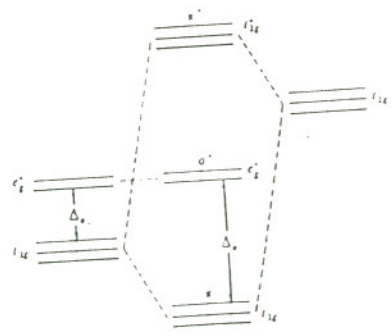
B


t_2
$2/5\ \Delta_t$
$3/5\ \Delta_t$
e

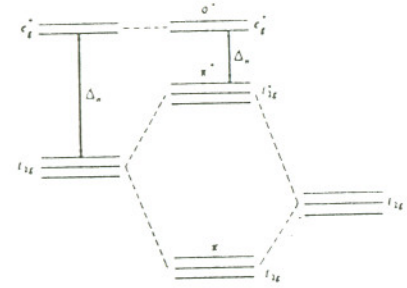
C


z^2
e_g
x^2-y^2
xz, yz
t_{2g}
xy

D


π^*
t_{1u}^*
t_{1u}
σ^*
e_g^*
e_g^*
Δ_o
Δ_o
t_{1u}
π
t_{1u}

E


σ^*
e_g^*
e_g^*
Δ_o
π^*
t_{1u}^*
Δ_o
t_{1u}
t_{1u}
π
t_{1u}

F